Data and Methodology Appendix

Technology Growth and Community Costs in Silicon Valley — California's 17th Congressional District (120th Congress boundaries, effective 3 January 2027). This appendix documents every dataset, processing decision, join, calculation and limitation behind the briefing. All numbers are produced by the scripts in `scripts/`; logs are in `logs/`.

A1. Dataset inventory and audit

Dataset (file)	Source / vintage	Geography & records	Identifier	Status in analysis
CA17_Study_Area.gpkg	Census 2026 TIGER/Line, 120th Congress CDs (layer tl_2026_06_cd120shp)	1 polygon, EPSG:4269	GEOID 0617	Used — study area
CA17_Census_Tracts.gpkg	Census 2026 TIGER/Line tracts (2020 tract vintage) selected with Intersects	202 polygons (Santa Clara 169, Alameda 32, San Mateo 1)	GEOID (11 char)	Used — overlap-classified (A3)
Median Household Income.zip	ACS 2020–2024 5-yr B19013 (data.census.gov)	88,605 rows, all U.S. summary levels → 9,129 CA tracts	GEO_ID 1400000US + 11	Used
Median Gross Rent.zip	ACS 2020–2024 5-yr B25064	same	GEO_ID	Used
Gross Rent as % of HH Income.zip	ACS 2020–2024 5-yr B25070 (001–011)	same	GEO_ID	Used
GrossRentas%ofincome.gpkg	Attribute-only copy of B25070 CSV (9,129 rows, no geometry)	—	GEO_ID	Not used (duplicate of zip)
calenviroscreen50results_f_070126.shp.zip	OEHHA CalEnviroScreen 5.0 final results (metadata 8 Jun 2026)	9,106 tracts, EPSG:3310; −999 = no data	tract (float, no leading 0)	Used
CaliforniaEA_ada.csv	CEC Energy Equity Indicators (energy affordability) — page states "Data is NOT final"; updated 15 Sep 2025	27,022 rows = 9,040 tracts × tenant type (COM 9,040 / REN 9,029 / OWN 8,953)	FIP (10 digits, no leading 0)	Joined; context only (draft)
Alameda Housing Data.xlsx; SC Housing Data.xlsx	HCD Housing Element Sites Inventory extract (land_use_last_update 2022-11-21)	17 parcels, all jurisdiction = City of Alameda	parcel_id / APN	Not used: files are byte-identical (MD5 e18ec5bf…) and cover the City of Alameda, outside CA-17
Ro Khanna Project.qgz	User's earlier QGIS project	references local TIGER zips + VersaTiles basemap	—	Inspected only
BAAQMD Permitted & Proposed Data Centers (Phase 1) FeatureServer; AGOL item 7d1bb8b5…	Bay Area Air District	—	—	NOT accessible: service HTTP 499 'Token Required'; item, /data and search return 403 / not indexed (anonymous). Not bypassed.
City of San José — Data Center and other Large Energy-Use Projects	sanjoseca.gov, page updated 11/04/2026; read 10 Nov 2026 in a browser (site blocks scripted fetches)	14 data-center + 2 large-energy records	SJPermits file #	Used — transcribed
CEC Power Plant Listing / SPPE dockets	energy.ca.gov, read 10 Nov 2026	Bay Area data-center backup-generation dockets	Docket #	Used — 6 records not on the San José list; 3 cross-referenced
OpenStreetMap via Overpass API	telecom=data_center or building=data_center, bbox 36.9–37.95N, 122.6–121.4W; OSM base 2026-11-11T01:12Z	85 features (5 flagged duplicates)	osm type/id	Used — existing facilities (context)
PNNL IM3 Open Source Data Center Atlas v2026.11.02	MSD-LIVE, doi:10.57931/2550666 (OSM-derived)	GPKG/CSV	id	Not downloaded: download requires MSD-LIVE CLI login / browser-issued credentials; same OSM source queried directly instead
Esri World Light Gray Canvas	server.arcgisonline.com tiles cached to GeoTIFF (≈9.55 m/px)	basemap	—	Used — cartography only

A2. Geographic processing

- All area calculations in EPSG:3310 (California Albers, NAD83, equal-area). Geometries validated (0 invalid) and passed through make_valid.
- Overlap share = area(tract ∩ CA-17) / area(tract). Sum of intersections equals the district area (490.73 km²), confirming complete tract coverage.
- Classes: within ≥90% (141), majority 50–90% (13), minority 5–50% (15), sliver <5% (33).
- **Analytical rule:** the district analysis set is the 154 tracts with ≥50% of area in CA-17 (Santa Clara 136, Alameda 18). Their 2024 ACS population (via CES 5.0 ACS2024Pop) is 754,222, close to the ≈760,350 equal-population target (2020 Census California population 39,538,223 ÷ 52 seats), so the rule reproduces the district population well. Minority and sliver tracts are retained in every layer (flag in_district_set = 0) and appear in regional analysis; nothing is dropped. Area weighting of counts was rejected because it assumes uniform population.

- County composition (overlay of CA-17 with county polygons dissolved from CES 5.0 tracts): Alameda 28.9% (140 km²); Santa Clara 71.1% (344 km²). No San Mateo County area; the single San Mateo tract (06081990100, water) is a boundary sliver. This matches the published Proposition 50 description of CA-17 (parts of Alameda and Santa Clara counties: Newark, part of Fremont; Cupertino, Milpitas, Santa Clara, Sunnyvale and part of San José).
- Regional frame: all 959 CES 5.0 tracts in Alameda, Santa Clara and San Mateo counties (2020 tract geometry, identical GEOIDs to TIGER 2020 tracts).
- Maps use Web Mercator (EPSG:3857) to match the basemap; scale bars are measured on the WGS84 ellipsoid. District maps clip tract polygons to the CA-17 boundary for display only.

A3. Variables and calculations

Field	Definition / formula
GEOID_std	11-character tract FIPS. ACS: last 11 characters of GEO_ID (prefix 1400000US). CES: int(tract) zero-padded to 11. CEC: FIP zero-padded. Original fields kept.
mhi, mhi_moe	B19013_001E/M, median household income (2024 dollars). '250,000+' → 250000 with mhi_flag = top-coded.
mgr, mgr_moe	B25064_001E/M, median gross rent. '3,500+' → 3500, top-coded flag. '-' → missing (suppressed / no renters).
rent_hh_computed	B25070_001 − B25070_011 (renter households for whom the rent-to-income ratio is computed). MOE = √(M001² + M011²).
pct_burden30	100 × (B25070_007 + 008 + 009 + 010) / rent_hh_computed. Numerator MOE = √ΣM². Proportion MOE: Census ACS Handbook (2020) formula √(MOEnum² − p²·MOEden²)/den, ratio formula if radicand < 0.
pct_burden50	100 × B25070_010 / rent_hh_computed (same MOE method).
*_cv, *_rel	CV = (MOE/1.645)/estimate × 100. Reliability: high ≤15, medium ≤30, fair ≤40, low >40. Maps hatch burden with CV >40 or <50 renter HH.
District pooled shares	Σ tract numerators / Σ tract denominators over the analysis set; MOEs via √ΣM² and the proportion formula. Medians are not pooled; the distribution of tract medians is reported.
ces_* fields	CES 5.0 overall score/percentile (CIscore/P), Pollution Burden (Pollution/S/P), Population Characteristics (PopChar/Sco/P) and indicator percentiles (statewide). −999 set to missing.
burden_t, ces_t, bivar	Terciles across the 959 regional tracts (burden only where rent_hh_computed ≥50). Cut points: burden 41.1% / 55.1%; CES 21.3 / 46.0 percentile. bivar = 'burden_t-ces_t'.
screen_overlap	1 if burden_t = 3 AND ces_t = 3 (no weighting, no index).
pop_1km_est	Σ tract population × share of tract area within 1 km of a facility. Screening approximation (uniform-population assumption).
osm_dc_n	Count of de-duplicated OSM data-center features within the tract (duplicates: same name within 150 m).

A4. Join validation and missing values

Join	Records	Matched	Unmatched	Unmatched IDs
tracts->ACS B19013/B25064/B25070	202	202	0	
tracts->CES5.0	202	200	2	06001990000;06081990100
tracts->CEC energy affordability	202	199	3	06001444303;06001990000;06081990100
region tracts (CES geom)->ACS	959	959	0	

- 06001990000 and 06081990100 are water-only tracts (0 population); CES 5.0 omits them. 06001444303 (within CA-17) has no households in ACS and no CEC record.
- Missing values (all 202 / district set): mhi 3/1; mgr 4/2; pct_burden30 3/1; pct_burden50 3/1; ces_CIscoreP 3/1; ces_PollutionP 2/0; ces_PopCharP 3/1; ea_REN_EBurden 3/1.
- Top-coding in the district set: 18 tracts at $250,000+ income and 40 at $3,500+ rent (listed in data/processed/acs_flags_ca17.csv).
- 12 district tracts have burden CV >40% (low reliability); 2 have <50 renter households and are left unclassified in the screen.

A5. CalEnviroScreen 5.0 — what it measures, and district results

CalEnviroScreen (OEHHA, for CalEPA) ranks census tracts by cumulative burden: Pollution Burden (exposure indicators such as ozone, PM2.5, diesel PM, traffic, toxic releases, drinking-water contaminants, lead risk, pesticides; and environmental-effects indicators such as cleanup sites, groundwater threats, hazardous-waste facilities, impaired waters, solid-waste sites) × Population Characteristics (asthma, low birth weight, cardiovascular disease; education, linguistic isolation, poverty, unemployment, housing burden). Scores are relative statewide percentiles. It does not measure facility-specific emissions, noise, heat, water withdrawals or electricity demand, and indicator data lag the release by one to several years. A high score therefore cannot be attributed to any individual facility.

Indicator (statewide pct.)	Median tract	Pop-weighted mean	% tracts ≥75th
gwthreatsP	76.8	70.3	52.6
trafficP	73.2	68.7	48.7
hazP	71.4	68.9	43.5
lingP	68.4	65.9	40.9
cleanupsP	66.7	61.2	45.5
dieselP	65.7	62.1	37.0
lbwP	59.7	57.3	24.0

Indicator (statewide pct.)	Median tract	Pop-weighted mean	% tracts ≥75th
SmATSP	57.9	57.7	31.8
drinkP	45.0	43.7	0.6
pmP	39.3	38.1	0.0
leadP	37.5	36.6	2.6
unempP	34.1	36.1	5.2
RSEIhazP	30.0	37.6	6.5
eduP	29.7	33.1	1.3
cvdP	22.8	23.0	0.0
housingBP	21.0	27.7	3.9
ozoneP	18.1	18.4	0.0
povP	16.8	21.7	0.6
iwbP	14.6	28.2	3.2
asthmaP	11.8	15.9	0.0
pestP	0.0	1.1	0.0
swisP	0.0	25.3	11.0

A6. Documented facility inventory (verified sources)

Each record is transcribed from an official web page (URL in data/external/inventory/facility_inventory_sources.csv), retains its permit or docket ID, and is geocoded with the U.S. Census Geocoder; five addresses not matched by Census were geocoded with OSM Nominatim (house-level except SJ14, street-level) and NorthTown (CEC04) is placed at the W Trimble Rd / Orchard Pkwy intersection named in its docket. Data centers and the City's 'large energy-use (not data center)' projects are kept separate. Out-of-region CEC dockets (RB Inyokern, Kern Co.; Vernon, Los Angeles Co.) were excluded; Pittsburg (Contra Costa) is kept but lies outside the map frame. 'Approved / exempted' means a planning approval or CEC exemption, not construction or operation.

ID	Project	Type	Source ID	Status	Submitted / approved	MW (basis)	Sq ft	In CA-17	Geocode
SJ01	150 S 1st Street data center (PD rezoning)	DC	PDC26-016	Proposed - under review	2026-09-10 /	—	—	No (0.7 km)	Census Geocoder
SJ02	326 Charcot Avenue data center	DC	SP26-026	Proposed - under review	2026-09-02 /	99 (CEC)	514,000	Yes	Census Geocoder
SJ03	5601 Great Oaks Parkway data center	DC	SP26-007	Proposed - under review	2026-08-10 /	—	444,436	No (11.9 km)	OSM Nominatim
SJ04	865 Embedded Way data center	DC	SP26-020	Proposed - under review	2026-08-03 /	18 (City)	152,676	No (10.8 km)	Census Geocoder
SJ05	5977 Silver Creek Valley Road data center	DC	SP26-016	Proposed - under review	2026-07-13 /	99 (City)	516,000	No (11.8 km)	Census Geocoder
SJ06	180 Park Avenue mixed-use / data center (Option B)	DC	PD25-015	Proposed - under review	2025-08-12 /	—	457,125	No (0.5 km)	Census Geocoder
SJ07	323 Terraine Street mixed-use / data center (Option B)	DC	PD25-011	Proposed - under review	2025-04-09 /	—	240,000	No (0.1 km)	Census Geocoder
SJ08	350 W Trimble data center	DC	CP23-013	Proposed - under review	2025-06-30 /	—	206,250	Yes	Census Geocoder
SJ09	300 S 1st Street mixed-use / data center	DC	PD25-013	Proposed - under review	2024-10-17 /	—	238,000	No (0.8 km)	Census Geocoder
SJ10	2549 Orchard Parkway data center (Microsoft SJDC 04)	DC	CP23-008	Proposed - under review	2022-11-15 /	97.8 (CEC)	630,000	Yes	OSM Nominatim
SJ11	300 Holger Way data center (conversion)	DC	SP25-029	Approved / exempted	2025-11-19 / 2026-07-08	—	97,800	Yes	Census Geocoder
SJ12	Fortune Drive / Trade Zone Blvd data centers	DC	PD22-001	Approved / exempted	2022-02-17 / 2023-05-09	91 (City)	522,194	Yes	Census Geocoder
SJ13	127 Great Oaks data center campus	DC	SP15-031	Approved / exempted	2015-07-01 / 2017-01-12	99 (City)	579,000	No (14.2 km)	Census Geocoder
SJ14	Microsoft San Jose Data Center (Zanker Rd / Alviso-Milpitas)	DC	CP23-016	Approved / exempted	2019-11-21 / 2025-04-09	99 (City)	397,205	Yes	OSM Nominatim
SJ15	5087 Disk Drive R&D; facilities	LEU (not DC)	H26-005	Proposed - under review	2026-03-06 /	250 (City)	482,790	Yes	Census Geocoder
SJ16	5853 Rue Ferrari R&D; facility	LEU (not DC)	HA21-006-01	Approved / exempted	2025-09-10 / 2025-12-10	49 (City)	300,000	No (13.2 km)	Census Geocoder
CEC01	Bowers Data Center	DC	22-SPPE-01	Approved / exempted	/	72 (CEC)	—	Yes	OSM Nominatim
CEC02	Lafayette Data Center	DC	20-SPPE-02	Approved / exempted	2020-05-20 / 2023-11-09	99.8 (CEC)	576,120	Yes	OSM Nominatim
CEC03	Martin Data Center	DC	22-SPPE-03	Approved / exempted	2022-11-17 /	96 (CEC)	—	Yes	Census Geocoder
CEC04	NorthTown Data Center	DC	25-SPPE-02	Proposed - under review	/	97.3 (CEC)	—	Yes	OSM street intersection
CEC05	STACK SVY03A Data Center Campus	DC	23-SPPE-01	Approved / exempted	/	76.6 (CEC)	—	No (10.1 km)	Census Geocoder
CEC06	AVAIO Pittsburg Data Hub	DC	24-SPPE-01	Approved / exempted	/	92 (CEC)	347,740	No (51.3 km)	Census Geocoder

- Cross-references: SJ02 = CEC 26-SPPE-03 (Charcot Ave); SJ05 = CEC 26-SPPE-02 (Silver Creek); SJ14 = CEC 19-SPPE-04 (Microsoft, 1657 Alviso-Milpitas Rd); SJ10 ≈ CEC 22-SPPE-02 (Microsoft SJDC 04; matched on site and building program). CEC04 NorthTown may relate to SJ08 (CP23-013) but acreage and generator counts differ, so both are retained.
- OpenStreetMap existing facilities: 85 features retrieved, 5 duplicates flagged (not deleted), 76 unique within the three-county frame, 63 inside CA-17. OSM is crowd-sourced; presence/absence is not verified and the layer must not be read as a census of facilities.
- The Bay Area Air District inventory could not be retrieved. If the Air District makes it public, scripts/02_facilities.py can ingest it as an additional layer, and maps re-export with scripts/05_build_qgis_project.py.

A7. Limitations

- Partial facility inventory (San José + CEC dockets + OSM). Cities of Santa Clara, Milpitas, Sunnyvale and Fremont publish no comparable list; projects below CEC thresholds outside San José are missing.
- ACS: 5-year pooled estimates (2020–2024); large tract MOEs; top-coding at $250,000 / $3,500 understates the upper tail; tract medians cannot be aggregated.
- Tract assignment by area share, not population; some residents of majority tracts live outside CA-17 and vice versa. Large, sparsely populated tracts (e.g., 5046.02 Alviso/baylands, 64 km², ≈1,500 residents) dominate visually.
- CalEnviroScreen is a relative screening tool; indicator vintages vary and pre-date recent projects. It does not capture backup-generator emissions specifically.
- Bivariate terciles are relative to the three-county region; different breaks would move tracts between classes. No composite index is constructed.
- MW figures mix bases (backup-generation capacity, maximum load, equipment capacity) and are not additive or indicative of actual consumption. CEC SPPE dockets cover 50–100 MW backup plants by definition, so the 90–99.8 MW clustering is partly a selection effect.
- CEC energy-affordability data are draft and were used only for a contextual statement (renter energy burden median ≈1.0% of income in district tracts; none above 6%).
- Correlation (Spearman rho = 0.36 between burden and CES across district tracts) and proximity are descriptive; no causal claim is made about data centers and housing costs or environmental scores.
- District boundary is the incoming 120th-Congress geography (effective 3 Jan 2027), not the 119th-Congress district currently in effect.

A8. Reproducibility

Environment: QGIS 3.44 bundled Python 3.12 (PyQGIS, GeoPandas, pyogrio, SciPy); openpyxl, ReportLab and pypdf installed to scripts/_pylib. Run from the project root:

```
source scripts/env.sh
$QGIS_APP/MacOS/python scripts/01_prepare_data.py   # geography, ACS, CES, CEC joins
$QGIS_APP/MacOS/python scripts/02_facilities.py     # geocode inventory, OSM, proximity
$QGIS_APP/MacOS/python scripts/03_summary_stats.py  # statistics -> summary_stats.json
bash scripts/04_cache_basemap.sh; $QGIS_APP/MacOS/python scripts/04b_city_labels.py
$QGIS_APP/MacOS/python scripts/05_build_qgis_project.py # QGIS project + 3 layouts + exports
$QGIS_APP/MacOS/python scripts/06_build_briefing.py; $QGIS_APP/MacOS/python scripts/07_build_appendix.py
```

Raw inputs in data/raw are read-only copies of the supplied files; external downloads (OSM JSON, geocode cache, transcribed inventory) are in data/external. The Overpass query and geocoder responses are cached so reruns are deterministic; delete the caches to refresh.

A9. Source citations (all accessed 10 November 2026 unless noted)

- U.S. Census Bureau. American Community Survey 2020–2024 5-Year Estimates, Tables B19013, B25064, B25070. data.census.gov.
- U.S. Census Bureau. 2026 TIGER/Line Shapefiles: Congressional Districts (120th Congress) and Census Tracts, California.
- U.S. Census Bureau (2020). Understanding and Using ACS Data: What All Data Users Need to Know (ACS Handbook) — MOE formulas.
- California Office of Environmental Health Hazard Assessment. CalEnviroScreen 5.0 results shapefile (CES5_final_shapefile, June 2026). oehha.ca.gov/calenviroscreen.
- City of San José, Planning, Building & Code Enforcement. “Data Center and other Large Energy-Use Projects” (updated 11/04/2026). sanjoseca.gov.
- California Energy Commission. Power Plant Listing; docket pages 19-SPPE-04, 20-SPPE-02, 22-SPPE-01, 22-SPPE-02, 22-SPPE-03, 23-SPPE-01, 24-SPPE-01, 25-SPPE-02, 26-SPPE-02, 26-SPPE-03. energy.ca.gov/proceedings/power-plant-listing. Lafayette address from CEC Draft EIR / CEQAnet SCH 2021080074.
- California Energy Commission. Energy Equity Indicators — Deep Dive: Energy (draft; data updated 15 Sep 2025). energy.ca.gov.
- Bay Area Air District. Data Centers and the Air District (baaqmd.gov/Rules-and-Compliance/Data-Centers); “Permitted and Proposed Data Centers Phase 1” FeatureServer (not publicly accessible).
- Mongird, K., Thurber, T., Vernon, C., Burleyson, C., Akdemir, K.Z., Rice, J. IM3 Open Source Data Center Atlas. PNNL / MSD-LIVE. doi:10.57931/2550666 (v2026.11.02).
- OpenStreetMap contributors (ODbL 1.0), via Overpass API (overpass-api.de) and Nominatim.
- California Senate Office of Demographics; California Secretary of State — Proposition 50 (2025) congressional map materials (county composition cross-check).
- Esri. World Light Gray Canvas Base and Reference (Esri, HERE, Garmin, © OpenStreetMap contributors).